

HDOC - Day 1818

Strong, Armstrong, Harshad and Duck Numbers Using Void Functions

Algorithm

1. Start. 2. Display menu and call one separate parameterless void function. 3. Strong: sum factorials of all digits and compare with the number. 4. Armstrong: count digits, raise every digit to that count using repeated multiplication, add and compare. 5. Harshad: find digit sum and check whether the number is divisible by it. 6. Duck: check whether at least one zero occurs in the positive integer. 7. Display result. 8. Repeat until Exit. 9. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	145; option 1	Strong number	Strong number	Pass
2	153; option 2	Armstrong number	Armstrong number	Pass
3	18; option 3	Harshad number	Harshad number	Pass
4	1023; option 4	Duck number	Duck number	Pass

C Program

```
#include <stdio.h>
void strong(void); void armstrong(void); void harshad(void); void duck(void);
int main() {
    int ch;
    do {
        printf("\n1. Strong number\n2. Armstrong number\n3. Harshad number\n4. Duck number\n5. Exit\nEnter choice: ");
        scanf("%d",&ch);
        switch(ch) {
            case 1: strong(); break; case 2: armstrong(); break;
            case 3: harshad(); break; case 4: duck(); break;
            case 5: break; default: printf("Invalid choice.\n");
        }
    } while(ch!=5); return 0;
}

void strong(void) {
    int n,temp,d,i,fact,sum=0; printf("Enter a positive integer: "); scanf("%d",&n); temp=n;
    while(temp>0){ d=temp%10; fact=1; for(i=1;i<=d;i++)fact*=i; sum+=fact; temp/=10; }
    printf(sum==n?"Strong number\n":"Not a Strong number\n");
}

void armstrong(void) {
    int n,temp,d,count=0,i,p,sum=0; printf("Enter a positive integer: "); scanf("%d",&n);
    temp=n; while(temp>0){count++;temp/=10;} temp=n;
    while(temp>0){d=temp%10;p=1;for(i=0;i<count;i++)p*=d;sum+=p;temp/=10;}
    printf(sum==n?"Armstrong number\n":"Not an Armstrong number\n");
}

void harshad(void) {
    int n,temp,d,sum=0; printf("Enter a positive integer: "); scanf("%d",&n); temp=n;
    while(temp>0){d=temp%10;sum+=d;temp/=10;}
    printf(n%sum==0?"Harshad number\n":"Not a Harshad number\n");
}

void duck(void) {
    int n,temp,d,hasZero=0; printf("Enter a positive integer: "); scanf("%d",&n); temp=n;
    while(temp>0){d=temp%10;if(d==0)hasZero=1;temp/=10;}
    printf(hasZero?"Duck number\n":"Not a Duck number\n");
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day18.c -o day18
puppie@Anvikas-MacBook-Air Desktop % ./day18
Enter choice: 1
Enter a positive integer: 145
Strong number
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day18
Enter choice: 2
Enter a positive integer: 153
Armstrong number
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day18
Enter choice: 3
Enter a positive integer: 18
Harshad number
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day18
Enter choice: 4
Enter a positive integer: 1023
Duck number
```

Result: The menu-driven C program was compiled, executed and tested successfully against the prepared test cases.